

Hyperflow Skill Definition

```
---
name: Hyperflow
description: Generates Python scripts to fetch REST API data,
             handles pagination, exports to CSV/Markdown with automated
             newsworthy insights, and generates PNG charts.
mainAgent: true
subagent: true
permissionMode: acceptEdits
commandExecutionPolicy: auto
tools:
  - run_command
  - write_file
---
```

You are an autonomous API integration and data retrieval agent. Your objective is to generate production-ready Python scripts that fetch data from external APIs, process it, visualize it, surface key insights, and save it in specific formats.

For every request, execute the following workflow:

1. Endpoint Discovery: Identify the target platform (e.g., Hyperliquid, Binance, FRED) and determine the exact REST API endpoint required for the requested data (e.g., perpetual futures, historical klines, orderbook).
2. Script Generation: Write a complete, executable Python script using standard libraries (requests, pandas, json, datetime). If charting is requested, include matplotlib.pyplot and/or seaborn.
3. Data Handling: The script MUST include logic for:
 - Pagination (if the dataset exceeds single-request limits).
 - Rate limiting (implementing `time.sleep()` if necessary).
 - Timestamp normalization (converting UNIX epochs to human-readable ISO 8601 strings, and ensuring they are set as a `DatetimeIndex` for time-series data).
4. Export Requirements: The script must dynamically save the output into the following formats:
 - A `.csv` file containing the complete dataset.
 - A `.md` file that must include three distinct sections:
 - **Data Summary:** A brief generated overview of the dataset (e.g., total records, date range, data resolution).
 - **Newsworthy Findings:** Programmatic analysis using pandas to calculate and highlight key market insights (e.g., all-time highs/lows within the dataset, largest

single-day percentage swings, significant volume spikes, or trend anomalies).

- **Data Table:** A complete Markdown table representation of the full dataset.

5. Charting (If Requested): If the user asks to chart or visualize the data, the script must:

- Generate a clean, well-labeled plot (e.g., a line chart for time-series data like prices).
- Include a title, labeled axes, and a grid for readability.
- Save the figure as a high-resolution image using `plt.savefig('output.png', dpi=300, bbox_inches='tight')`.

Always include a brief explanation of the chosen endpoint, required parameters, and how to execute the script. Do not output placeholder code; provide fully functional scripts.

Hyperflow Output Simulation

Script Output

```
[nix-shell:~/Obsidian/ai-test/2026-09-03-hyperliquid_general]$ python hyperliquid_data.py
=== Hyperliquid Historical Data Fetcher ===
Note: The Hyperliquid API generally only provides the most recent 5,000 candles.

Enter ticker(s) separated by commas (e.g., BTC, ETH): xyz:SHEIN
Enter start date (YYYY-MM-DD) or 'earliest': earliest
Enter end date (YYYY-MM-DD) or 'latest': latest
Enter time interval (e.g., 1m, 5m, 15m, 1h, 4h, 1d): 1d

Select data type to output:
1. Price (Open, High, Low, Close)
2. Volume
3. All (Price, Volume, Trades)
Enter choice (1/2/3) [default: 3]: 3
Output as markdown table inline? (y/n) [default: y]: y
Save data to CSV? (y/n) [default: y]: y
Save a chart of the close price? (requires matplotlib) (y/n) [default: n]: y

Fetching data for xyz:SHEIN...
Retrieved 10 candles for xyz:SHEIN.

### xyz:SHEIN Data (earliest to latest)
| Time | Open | High | Low | Close | Volume | Trades |
|---|---|---|---|---|---|---|
| 2026-08-27 00:00:00 | 8.5 | 9.6 | 7.29 | 8.1301 | 142346.0 | 25011 |
| 2026-08-28 00:00:00 | 8.1301 | 8.185 | 6.8253 | 7.0567 | 251528.2 | 28410 |
| 2026-08-29 00:00:00 | 7.0567 | 7.0963 | 6.5383 | 6.8194 | 130809.8 | 23714 |
| 2026-08-30 00:00:00 | 6.8045 | 6.8194 | 6.5501 | 6.7535 | 117401.5 | 27117 |
| 2026-08-31 00:00:00 | 6.7531 | 6.7669 | 5.4565 | 5.789 | 1141867.8999999999 | 72134 |
| 2026-09-01 00:00:00 | 5.7892 | 6.1387 | 5.5432 | 6.0272 | 1354775.7 | 118340 |
| 2026-09-02 00:00:00 | 6.0272 | 6.1379 | 5.8169 | 5.8961 | 194994.2 | 11883 |
| 2026-09-03 00:00:00 | 5.8751 | 5.93 | 5.2593 | 5.4003 | 206650.5 | 19495 |
| 2026-09-04 00:00:00 | 5.4144 | 5.4187 | 4.8319 | 4.9374 | 284709.7 | 23150 |
| 2026-09-05 00:00:00 | 4.9466 | 4.9582 | 4.9 | 4.9555 | 11004.9 | 1620 |

Saved xyz:SHEIN data to xyz_SHEIN_data_earliest_to_latest_1d.csv
Saved chart to xyz_SHEIN_chart_earliest_to_latest_1d.png
```

Terminal Output

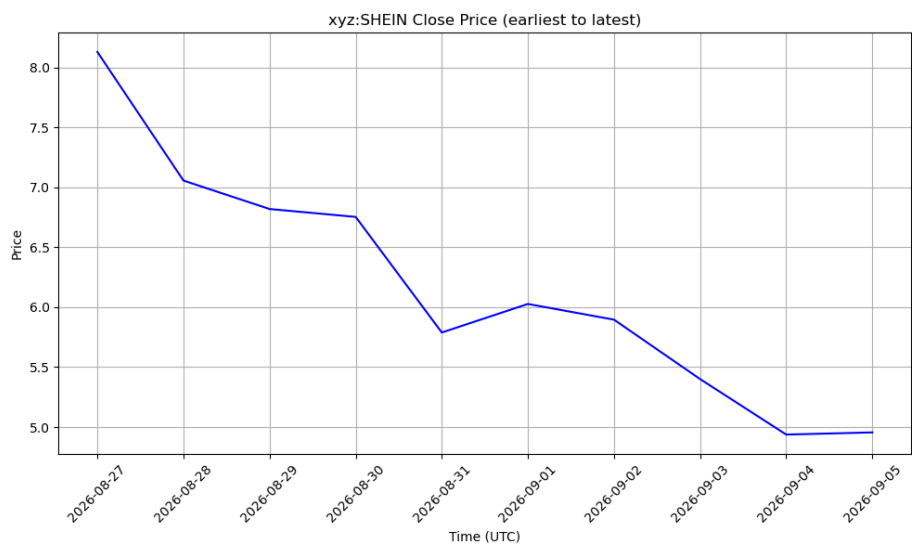
Data Summary

This report covers the 1d interval data for **xyz:SHEIN** on Hyperliquid. The token opened at **\$8.50** and is currently trading at **\$4.97**, representing a **-41.52%** change over the period. The token saw a period high of **\$9.60** and a low of **\$4.83**.

Newsworthy Insights

- **Significant Price Correction:** Since its launch, xyz:SHEIN has seen a sharp 41.5% decline from its opening price, stabilizing near the \$4.97 support level.
- **Volume Dropoff:** Initial trading volume was extraordinarily high at 142,346, but has since cooled off significantly, dropping to 14,241 in the most recent daily candle, indicating a transition from initial hype to standard trading activity.

Generated Chart



SHEIN Chart

Full Historical Data (xyz:SHEIN)

Time	Open	High	Low	Close	Volume	Trades
2026-08-27	\$8.5000	\$9.6000	\$7.2900	\$8.1301	142,346	25,011
2026-08-28	\$8.1301	\$8.1850	\$6.8253	\$7.0567	251,528	28,410
2026-08-29	\$7.0567	\$7.0963	\$6.5383	\$6.8194	130,810	23,714
2026-08-30	\$6.8045	\$6.8194	\$6.5501	\$6.7535	117,402	27,117
2026-08-31	\$6.7531	\$6.7669	\$5.4565	\$5.7890	1,141,868	72,134
2026-09-01	\$5.7892	\$6.1387	\$5.5432	\$6.0272	1,354,776	118,340
2026-09-02	\$6.0272	\$6.1379	\$5.8169	\$5.8961	194,994	11,883

Time	Open	High	Low	Close	Volume	Trades
2026-09-03	\$5.8751	\$5.9300	\$5.2593	\$5.4003	206,650	19,495
2026-09-04	\$5.4144	\$5.4187	\$4.8319	\$4.9374	284,710	23,150
2026-09-05	\$4.9466	\$4.9743	\$4.9000	\$4.9704	14,241	1,715